

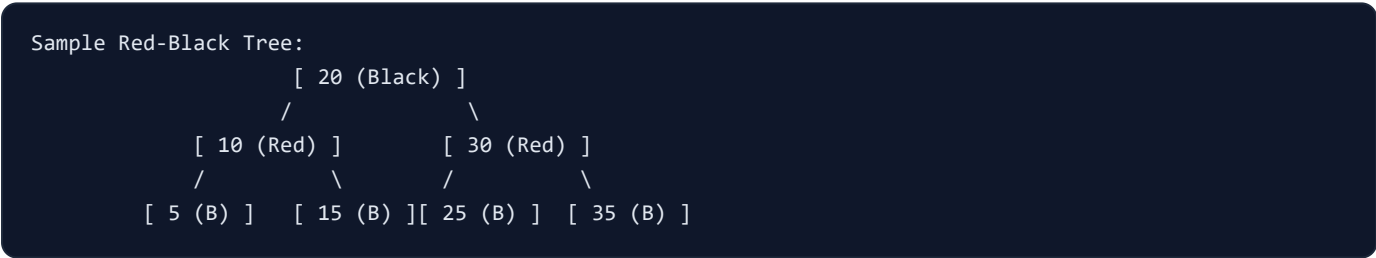
Self-Learning: Red-Black Trees — Data Structures

Definition: A **Red-Black Tree** is a self-balancing Binary Search Tree where each node stores an extra bit representing **color (Red or Black)**, satisfying specific balance constraints that ensure the tree height remains bounded by $2 \lceil \log_2(N + 1) \rceil$.

1. Detailed Technical Explanation

The 5 Red-Black Tree Properties:

- 1. **Node Color:** Every node is either **RED** or **BLACK**.
- 2. **Root Property:** The root node is always **BLACK**.
- 3. **Leaf Property:** Every leaf (**NIL** / **NULL** sentinel node) is **BLACK**.
- 4. **Red Property:** If a node is **RED**, both of its children must be **BLACK** (No two consecutive RED nodes on any path).
- 5. **Black-Height Property:** For every node, every simple path from that node to any of its descendant **NIL** leaves contains the **exact same number of black nodes** (Black-Height).



2. AVL Tree vs Red-Black Tree Comparison

Feature	AVL Tree	Red-Black Tree
Balance Strictness	Strictly Balanced (**)	BF
Search Speed	Faster lookups due to strictly smaller height.	Slightly slower search than AVL.
Insertion / Deletion Speed	Slower (frequent rebalancing rotations).	Faster insertions/deletions (at most 2-3 rotations).
Use Cases	Search-intensive workloads (Lookups > Writes).	Standard library maps (C++ <code>std::map</code> , Java <code>TreeMap</code>).

3. Insertion in Red-Black Tree

When inserting a new key:

1. Insert as standard BST leaf and color it **RED**.
 2. If root: Color **BLACK**.
 3. If parent is **RED** (Double-Red Violation):
 - **Case 1: Uncle is RED:** Perform **Color Flip** (Parent & Uncle -> Black, Grandparent -> Red) and propagate up.
 - **Case 2: Uncle is BLACK (Triangle - LR/RL):** Perform single rotation on parent to convert into Line.
 - **Case 3: Uncle is BLACK (Line - LL/RR):** Perform rotation on Grandparent and swap colors of Parent and Grandparent.
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4. Quick Recall Flow

RB-Tree -> 5 Rules: Root Black, Leaves Black, No Consecutive Reds, Uniform Black-Height -> Color Flips & Rotations